Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/31/2007
PI Name:	Zhong, Katie Wei Hong Ph.D.		
Project Title:	Hybrid Composites with Reactive Nano-matrix for Cosmic Radiation Shielding		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pullman	**State:**	WA
Zip Code:	99164-2920	**Congressional District:**	5
Comments:	Formerly at North Dakota State University; moved September 2007 to Washington State University.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	07/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:	NOTE: PI moved to Washington State University from North Dakota State University in August 2007.		
COI Name (Institution):	Stone, James (North Dakota State University)		
Grant/Contract No.:	NNM04AA62G		
Performance Goal No.:			
Performance Goal Text:			
	The goal of this project is to design and manufacture more reliable multifunctional high performance shielding materials i.e., ultrahigh molecular weight polyethylene (UHMWPE) fiber /graphite fiber (CF) hybrid composites with reactive nano-matrix. The novel polymer nano-matrices with reactive molecular structures can effectively improve the adhesion and bonding strength of UHMWPE fiber and CF to the nano-matrix. The resulting UHMWPE fiber/CF hybrid composites will achieve significant enhancement of structural integrity and safety, and maintain excellent cosmic radiation shielding properties. Instead of a simple mixture of nanofibers and polymers to create a nanocomposite, the nano-matrix materials have unified molecules or carbon nanofiber-thermoset resin pendant reactive groups. Due to functions of reactive graphitic nanofibers (r-GNFs) with linker molecules containing reactive amino groups, the nano-matrix can not only have		

Task Description:

drastically enhanced mechanical properties, but can also obtain improved interface adhesion between the continuous fibers and the polymer nano-matrix without high costs in surface treatment for the reinforcements. Moreover, since much lower nanofiber-loading is involved in the process, changes of current manufacturing procedures of shielding composite structures used in NASA missions are not required. Therefore, composites containing UHMWPE fiber and the super nano-matrix are cost-effective, and will possess excellent cosmic radiation shielding performance and enhanced structural reliability and safety.

The proposed research activities include: (a) Fabrication of reactive nano-matrix materials. Based on derivatized GNFs, the reactive nano-matrix containing r-GNFs-epoxy with controlled nanofiber size and low fiber loading will be fabricated. (b) Structure and Property Characterization of Nano-matrices. The micro- and nano-scale structures, curing mechanisms, mechanical properties and aging resistance of the nano-epoxy matrix will be investigated. (c) Study of interface adhesion of UHMWPE fiber to the nano-matrices. The contact angle of the nano-epoxy matrix to the UHMWPE material. (d) Fabrication of UHMWPE fiber reinforced nano-matrix composites and UHMWPE/graphite hybrid fibers reinforced nano-matrix composites, and investigation of mechanical properties. (e) Characterization of shielding performance in NASA Space Radiation Laboratory (NSRL), study of the effect of the irradiation on reliability and safety issues for the composites.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

1. Summary of achievements
We have conducted successful research with this NASA grant. Our achievements include:

Peer review journal papers: 19 (another 5 are in review) Theses/Dissertations: 3 Conference papers: 17 Invited talks/seminars: 10 Awards: 2 Grant from an Aero company: 1 Courses developed: 2

2. Broad impact

The results of this research project have gained significant attention by several industrial companies including the aerospace companies Boeing Co., (Seattle) and Spirit Aerosystems Inc. (Wichita), and with leading composites and nanomaterial suppliers such as Hexcel Co., Toho Industries In., Toray (Japan) Co. and nanomaterials company such as the Zyvex Co. in Dallas.

Based on the work performed at NDSU through this NASA support, I was invited to make 10 talks/seminars on nanocomposites in industry (including the Boeing Co.) and universities. Currently, I am a consultant of The Boeing Company in the area of nanotechnology. The Boeing Co. has expressed strong interests in nanotechnology projects that the I plan to work on for applications in its next generation commercial airplanes. A significant research project on composite materials with nano-epoxy matrix for application in new generation commercial airplanes has been awarded to me by the Spirit Aerosystems Inc. Both the Hexcel Co and Zyvex Co have provided no-charge materials to me for related research and I have signed PIAs with Toho Composites and received a Letter of Support from the Toray co. based on this work.

I served as a reviewer and a panelist for reviewing NSF grant proposals and reviewed over ten journals on nanocomposites and nanotechnologies due to conducting this successful research project.

I developed and taught a class “Polymer Nanocomposites and Functionalities” for the Boeing Co. (30 hours; 60 engineers; twice) and a course “Nanocomposites” for NDSU graduate students (40 hours; 10 graduate students; once) based on the studies mainly supported by this NASA grant.

Through conducting this project, we developed the following two innovative technologies:

1. “Fiber Reinforced Reactive Nano-epoxy Composites” 2. “Improvement in Wetting and Adhesion by Modifying Polymer Resin Materials”

One of the impacts and potential benefits of these technologies is in extending the application space for the UHMWPE fibers in aeropace and airplane structures. UHWMPE fibers have comprehensive and attractive advanced properties including ultralight weight, high strength, high modulus, and cosmic radiation resistance and anti-corrosion properties. However, the weak interfacial adhesion with polymer resins is the inherent problem when this kind of fiber is used as a reinforcement in composite materials. Our research created a foundation for the development and design of composites reinforced with UHMWPE fiber, and will increase the scope of their applications to more complex, highly loaded space and aero structures. It is important to note that this technlogy is a cost effective approach with no impact to current production, and in fact reduces current processing steps. And most importantly, this class of composites can offer an alternative to the widespread use of carbon composites with can cause galvanic corrosion to contiguous aluminum, requiring heavy and costly barriers in aerospace applications. Carbon fibers additionally are subject to threats of global shortages as non-aerospace applications increase, so alternative UHMWPE fiber composites offer a risk reduction strategy for aerospace end-users if the deficiencies associated with weak interfacial adhesion has been solved.

Through over 3-year studies, all the research tasks have been completed.
1. A nano-epoxy matrix material with enhanced properties for UHMWPE fiber composites was developed through solving the challenging problems:

(1) Application of functionalized graphitic nanofibers (GNFs), and then improved the interphase between nanofibers/epoxy.

(2) Realization of chemical bonding between graphitic nanofibers and epoxy resin molecules through developing reactive nanofibers (r-GNFs).

(3) Realization of uniform dispersion of the nanofibers in epoxy resin.

(4) Examination of the effects of the nanofibers on properties of the nano-epoxy matrix, and determination of the optimized parameters for curing mechanism

Task Progress:	2. The mechanical and thermal properties as well as other properties of the nano-epoxy as a matrix for composite materials were characterized: (1) Mechanical: Flexural strength, modulus and breaking strain by three-point bending tests; fracture toughness by single edge notched bending tests; (2) Thermal: storage modulus E' and glass transition temperature Tg by DMA; CTE by thermal mechanical analysis (TMA) (3) Relaxation behavior: differential scanning calorimetry (DSC) (4) Viscosity (static): viscometer (5) Chemorheological: Dynamic viscosity, gelation time, Activation energy 3. The wettability of the nano-epoxy matrix to UHMWPE fibers through using the methods was studied: (1) Contact angle analysis (2) Drop spreading test 4. Interfacial adhesion properties in UHMWPE fiber/nano-epoxy composites were investigated: (1) Bundle composites Pullout test (2) Microbond test 5. The UHMWPE fiber/nano-epoxy composites in bundle form were studied: (1) Tensile at room temperature. (2) Effects of hygrothermal and UV on Tensile property 6. We fabricated various composite laminates with the nano-epoxy resins: three kinds of carbon fibers reinforced nano-epoxy composites, glass fiber reinforced nano-epoxy composites, and Kevlar fiber reinforced nano-composites for mechanical tests. We fabricated the UHMWPE/CF hybrid fibers reinforced nano-matrix composites. We studied interlaminar shear properties of the unidirectional composite laminates. 7. UHMWPE fiber/nano-epoxy composites and hybrid fibers reinforced composites were prepared for cosmic radiation tests. The cosmic radiation tests have been completed by Dr. Jack Miller (consultant of this NASA project) and his group at Brook Haven National Laboratory.

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